

Integer Data Type and if-else condition

Priority	Operators	Associativity
1 (Highest)	()	Left to Right
2	* / %	Left to Right
3	+ -	Left to Right

<pre>int temp; temp = a; a = b; b = temp;</pre>	<pre>a = a + b; b = a - b; a = a - b;</pre>
<pre>a = a * b; b = a / b; a = a / b;</pre>	<pre>a = a ^ b; b = a ^ b; a = a ^ b;</pre>
<pre>if (condition) { // statements executed when condition is TRUE } else { // statements executed when condition is FALSE }</pre>	<pre>if (condition1) { // statements } else if (condition2) { // statements } else if (condition3) { // statements } else { // statements }</pre>
<pre>if (condition1 && condition2) { // Both conditions are true } else { // One or both conditions are false }</pre>	<pre>if (condition1 condition2) { // One or both conditions are true } else { // Both conditions are false }</pre>
<pre>if ((condition1 && condition2) condition3) { // statements } else { // statements }</pre>	

Attendance %	Marks Awarded
80 – 85%	1
86 – 90%	2
91 – 95%	3
Above 95%	5

```
#include <stdio.h>
void main() {
    int attendance=90;
    int marks;
    if (attendance >= 80 && attendance <= 85) {
        marks = 1;
    } else if (attendance >= 86 && attendance <= 90) {
        marks = 2;
    } else if (attendance >= 91 && attendance <= 95) {
        marks = 3;
    } else if (attendance > 95) {
        marks = 5;
    } else {
        marks = 0;
    }
    printf("Marks Awarded: %d", marks);
}
```

```
#include <stdio.h>
void main() {
    int attendance=97;
    int marks;
    if (attendance > 95) {
        marks = 5;
    } else if (attendance >= 91) {
        marks = 3;
```

```
} else if (attendance >= 86) {  
    marks = 2;  
} else if (attendance >= 80) {  
    marks = 1;  
} else {  
    marks = 0;  
}  
printf("Marks Awarded: %d", marks);  
}
```

Challenge 2

Odd or Even

```
#include <stdio.h>  
void main() {  
    int num = 15;  
    if (num % 2 == 0)  
        printf("Even\n");  
    else  
        printf("Odd\n");  
}
```

Negative or Positive

```
#include <stdio.h>  
void main() {  
    int num = -8;  
    if (num < 0)  
        printf("Negative\n");  
    else  
        printf("Positive\n");  
}
```

Divisibility Check by 3 and 5

```
#include <stdio.h>
```

```
void main() {  
    int num = 30; // change this value to test  
    if (num % 3 == 0 && num % 5 == 0)  
        printf("Divisible by both 3 and 5\n");  
    else  
        printf("Not divisible by both 3 and 5\n");  
}
```

Leap Year Checker

```
#include <stdio.h>  
void main() {  
    int year = 2024;  
    if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))  
        printf("Leap Year\n");  
    else  
        printf("Not a Leap Year\n");  
}
```

Swap first and last digit of a 3-digit number

```
#include <stdio.h>  
void main() {  
    int num = 123;  
    int first = num / 100;    // gets the hundreds digit  
    int last = num % 10;      // gets the units digit  
    int middle = (num / 10) % 10; // gets the tens digit  
    int result = (last * 100) + (middle * 10) + first;  
    printf("Original: %d\n", num);  
    printf("Swapped: %d\n", result);  
}
```

Original: 123

Swapped: 321

Reverse a 3-digit number

```
#include <stdio.h>  
void main() {  
    int num = 123; // change this value to test
```

```
int first = num / 100;    // hundreds digit
int middle = (num / 10) % 10; // tens digit
int last = num % 10;      // units digit
int reversed = (last * 100) + (middle * 10) + first;
printf("Original: %d\n", num);
printf("Reversed: %d\n", reversed);
}
```

Original: 456

Reversed: 654

Remove the last digit from the given number

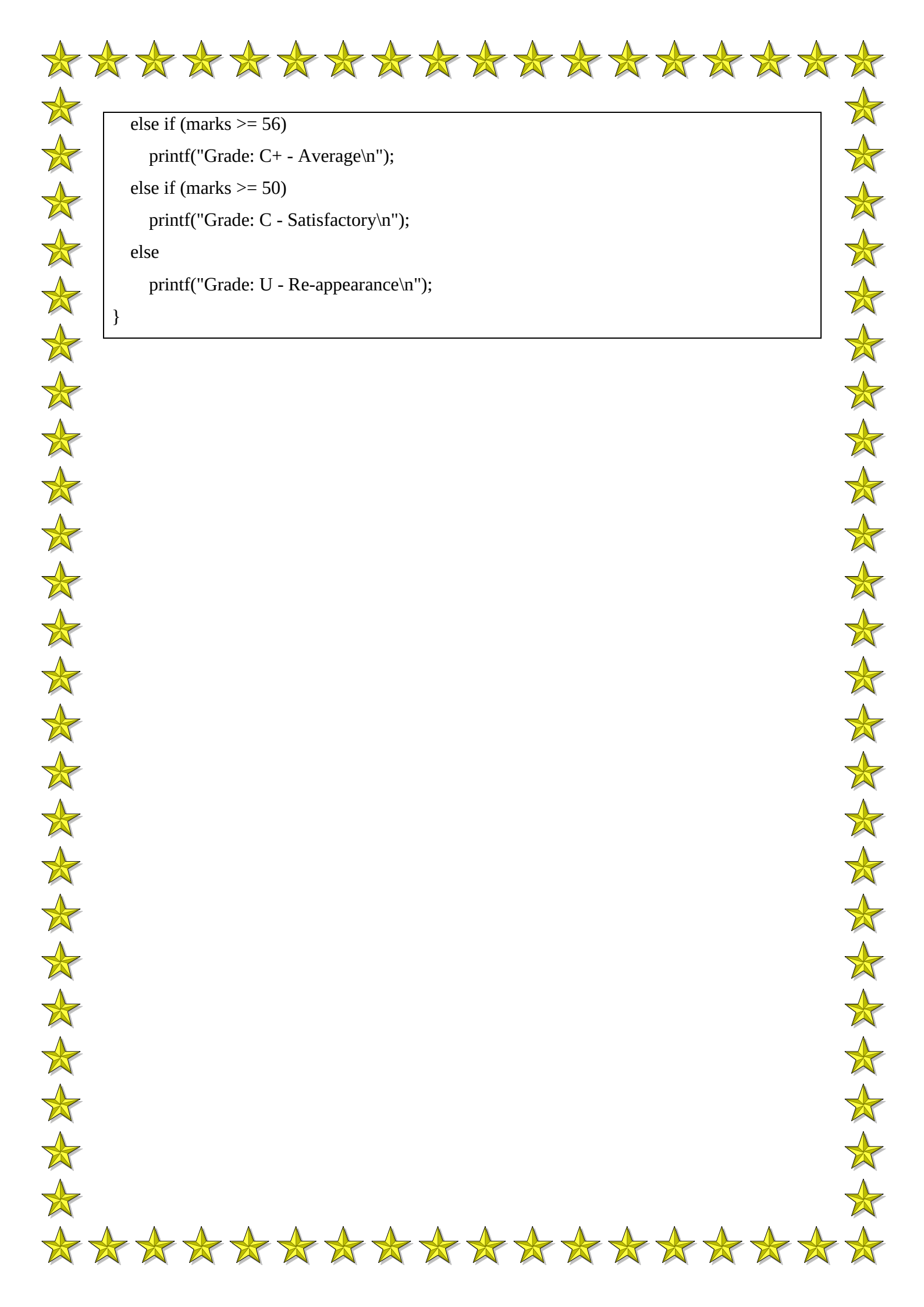
```
#include <stdio.h>
void main() {
    int num = 4326;
    int result = num / 10;
    printf("Original: %d\n", num);
    printf("After removing last digit: %d\n", result);
}
```

Original: 4326

After removing last digit: 432

Grade Problem

```
#include <stdio.h>
void main() {
    int marks = 78;
    if (marks >= 91)
        printf("Grade: S - Outstanding\n");
    else if (marks >= 81)
        printf("Grade: A+ - Excellent\n");
    else if (marks >= 71)
        printf("Grade: A - Very Good\n");
    else if (marks >= 66)
        printf("Grade: B+ - Good\n");
    else if (marks >= 61)
        printf("Grade: B - Above Average\n");
}
```



```
else if (marks >= 56)
    printf("Grade: C+ - Average\n");
else if (marks >= 50)
    printf("Grade: C - Satisfactory\n");
else
    printf("Grade: U - Re-appearance\n");
}
```